Management Information Systems

Week 1 Part II

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Weekly Topics

- **Subject to change without prior notice**
- **Ad-hoc in-class writing or quiz and discussions will be given**

W	Topic	W	Topic
1	Orientation **Introduction to MIS**	9	Telecommunications, the internet, and wireless technology
2	Information Systems in global business today	10	Securing information systems
3	Global E-business and Collaboration	11	*No Class*
4	Information Systems, Organizations, and strategy	12	Achieving operational excellence and customer intimacy
5	Ethical and Social issues in information systems	13	E-commerce: digital markets, digital goods
6	IT infrastructure and emerging technologies	14	Managing knowledge
7	Foundations of Business Intelligence	15	Enhancing decision making / Building information systems
8	*Midterm Exam*	*16*	*Final Exam*

Week 1

Pt. II Introduction to MIS

Laudon, K. C., & Laudon, J. P. (2022). *Management Information Systems: Managing the Digital Firm* (16th ed.). Pearson.
Simon, H. A. (1977). *The New Science of Management Decision*. Prentice Hall.
Ackoff, R. L. (1989). From data to wisdom. *Journal of Applied Systems Analysis*, 16, 3–9.
Sharda, R., Delen, D., & Turban, E. (2023). *Analytics, Data Science, & Artificial Intelligence*. Pearson.

Questions to Answer

1. **Why is information important, and what purposes does it serve?**

2. **How is information distinguished from data and knowledge?**

3. **What is Management Information Systems (MIS), and why should we study it?**

4. **What happens when critical information is missing in decision-making?**

5. **In the age of AI, who makes decisions: humans, AI, or both?**

0-1.

Why is information important, and what purposes does it serve?

Warm-up Question

- **Question**
  - Why do good managers sometimes make bad decisions?
    (Why do organizations with talented people still make poor decisions?)

- **Common answers**
  - Lack of ability
  - Lack of experience
  - Time pressure
  - Emotional state

Key Hypothesis

- **Many poor decisions are not caused by people, but by _____.**
  - Same problem, Same capability
  - Different _____ → Different decisions

Scenario & Task

- **Scenario: You are an operations manager of an online subscription service.**
  - Poor financial performance compared to last month
  - You must propose an action plan within one week

- **Task**
  - Working only with the information provided, identify the main problem
  - Prepare one clear decision (i.e., what should we do?)

#1

- **Based *only* on this data, what information can you extract, and what decisions would you make based on it?**

Metric	Last Month	This Month
Marketing Expenditure	$120,000	$145,000
Website Visitors	480,000	500,000
Conversion Rate	3.2%	2.6%
New Subscribers	15,360	13,000

#2

- **Based *only* on this data, what information can you extract, and what decisions would you make based on it?**

Metric	Last Month	This Month
Active Subscribers	210,000	205,000
Monthly Churn Rate	3.5%	5.2%
Repeat Subscription Rate	68%	59%
Avg. Subscription Length	11.2 months	9.1 months

#3

- **Based *only* on this data, what information can you extract, and what decisions would you make based on it?**

Metric	Last Month	This Month
Avg. Customer Support Response Time	6 hrs	14 hrs
Payment Failure Rate	1.8%	4.9%
Service Downtime (hrs/month)	1.2	5.6
Customer Complaints	820	1,450

Key Insight

- **The same problem can lead to different decisions**
- **Decision depends on *what information is visible***
- **Data must be relevant, *structured*, and decision-oriented**

Information Shown	Likely Decision
Marketing data	Cut or optimize advertising
Customer behavior data	Improve retention
Operations data	Fix systems and processes

Which decision is correct?
Or are all of them reasonable?

What additional information would you ask for before making a final decision?

0-2.

How is information distinguished from data and knowledge?

Data vs. Information vs. Knowledge

- **Data:** raw facts, numbers, records
- **Information:** processed and organized data
- **Knowledge:** understanding and experience

- **MIS focuses on transforming data into *actionable* information.**

The Knowledge Value Chain: Data

- **Data has in itself little semantic content**

- **Data has more meaning when presented as part of a structure**

Prod No.	Type	Make	Model	Price($)	Code
4825	Wagon	Ford	1994	5000	Almad21

Source: Minder Chen, IBM Academic Program course materials

The Knowledge Value Chain: Information

- **Information is relevant if it makes a difference to receive it or not**

Prod No.	Type	Make	Model	Price($)	Code
4826	Car	**BMW 328**	**1997**	5000	Almad21

Consumer Report
Price of Cars: BMW328

1996	**$12,000**
1997	$19,000
1998	$27,000
1999	$32,000

Source: Minder Chen, IBM Academic Program course materials

The Knowledge Value Chain: Knowledge

- **Information is the source of knowledge**

Example: Loan Risk Assessment

Name	Client Since	Credit E valuation	Assets	Loan		Risk
Rich Harris	1985	Good	20,000	20,000	…	Medium
John Leggo	1997	Fair	100,000	130,000	…	High
Sandra Hilt	1992	Good	10,200	2000	…	Low
…	…	…	…	…	…	…

Knowledge (as a business rule)
"*if we know the applicant, and his/her credit is good, and the applicant has assets for at least twice the amount of the loan,*
Then the risk is low, and unless you find something really strange, the loan should be granted"

Source: Minder Chen, IBM Academic Program course materials

0-3.

What is Management Information Systems, and why should we study it?

What is MIS?

- **Management Information Systems (MIS) is about…**
  - How organizations use information
  - How systems support managerial decision-making
  - How technology enables organizational performance

- **MIS is…**
  - Systems that provide ***information*** to support ***managerial decision-making*** and organizational control.

Management	Information	System

Management

- **Management is the effective utilization of human and material resources to achieve the enterprise objective**

- **It is a process consisting of the five basic functions:**
  - Planning
  - Organizing
  - Staffing
  - Directing
  - Controlling

Information

- **Information means the *processed data* that helps the management in planning, controlling and operations**

- **Data means *unstructured raw facts*, observations or unevaluated messages in isolation.**

- **Data involves facts and figures. Information on the other hand is like a finished product.**

System

- **A system is an *orderly grouping* of interdependent components linked together according to a plan to achieve a specific objective**
  - Human body is a system composed of various parts, which are working together towards a common objective

- **Information system:** an organized combination of people, hardware, software, communications network, and data resources that collects, transforms, and disseminates in an organization
  - Components: input, processor, output, people, hardware, software, data, network

MIS

- **An MIS is..**
  - An integrated user-machine system
    for providing information
    to support the operations, management, analysis, and decision-making functions
    in an organization

  - The system utilizes
    - Computer hardware and software
    - Manual procedures
    - Models for analysis, planning, control and decision making and
    - A database

MIS Is Not Just About IT

- **MIS is NOT about…**
  - Programming
  - Learning software tools
  - Databases only

- **Instead, MIS integrates people, processes, data, and technology to support _____.**

Why Study MIS?

- **Modern organizations operate in environments characterized by:**
  - High uncertainty
  - Rapid technological change
  - Data-intensive decision-making

  ➔ Understanding MIS is essential for managing complexity.

- **Doesn’t AI have all the information required to make the optimized decision? ➔ ?**

0-4.

What happens when critical information is missing in decision-making?

Organizations as Decision Systems

- **Organizations can be viewed as:**
  - Collections of resources
  - Sets of processes
  - ***Systems of decisions***

- **Managers do the following:**
  - Identify problems
  - Allocate resources
  - Monitor performance
  - Make strategic and operational ***decisions***

- **MIS shapes how decisions are made, repeated, and improved**

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ECONOMY • INFLATION

‘This is a wacky number’: economists cry foul as new government data assumes zero housing inflation in surprising November drop

By **Eva Roytburg**
Fellow, News

December 18, 2025, 2:01 PM ET

Add us on

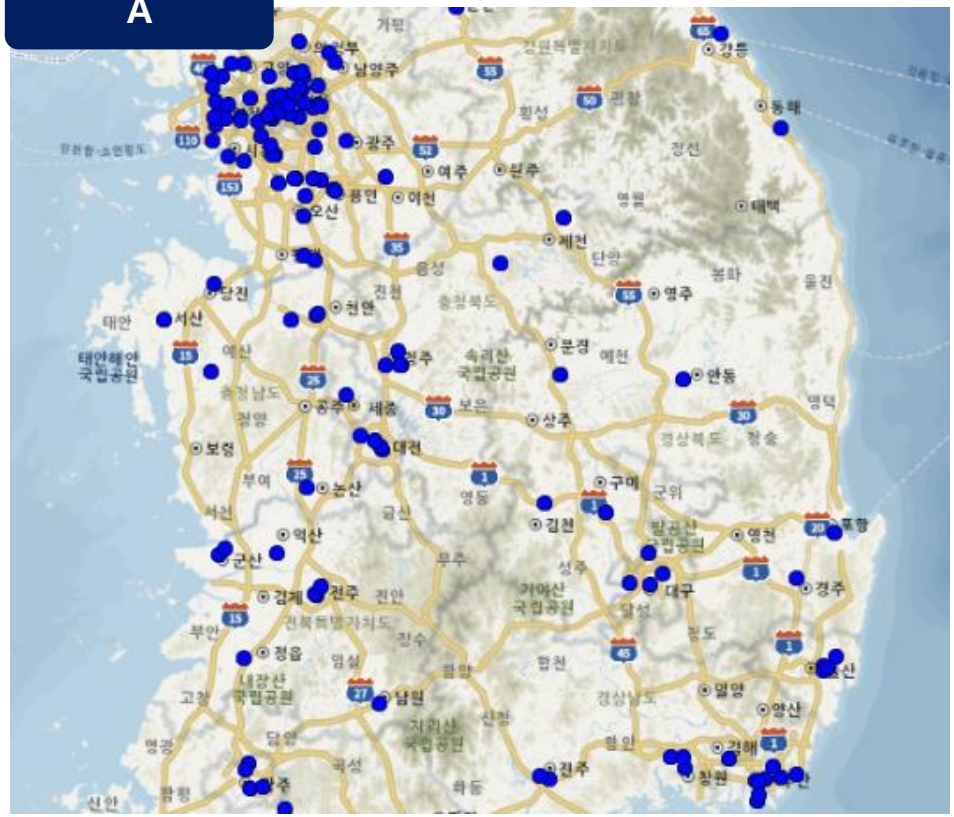
President Donald Trump pinned the country's economic woes on th
Dec. 17.
DOUG MILLS - POOL/GETTY IMAGES

https://fortune.com/2025/12/18/inflation-report-october-2025-shutdown-distorted-diane-swonk

Decision Support Systems

- **What do you think the main purpose of building this system is?**
- **What kind of decision users want to support using this system?**

A



B

0-5.

In the age of AI, who makes decisions: humans, AI, or both?

Video

- **Humans vs. AI: who should make the decision?**

Summary of the Video

- AI outperforms humans when it has high confidence, while humans perform better in uncertain or complex cases by using context
- The best approach is augmented intelligence: combining AI recommendations with human judgment achieves the highest overall decision accuracy across many situations.
- Human cognitive bias can undermine AI support: forced display of AI recommendations can lead to automation bias, causing humans to over-trust AI and ignore their own judgment.
- How AI advice is presented matters: optional display and careful presentation of AI recommendations reduce bias and improve trust, enabling humans and AI to work effectively as a team.

Discussion

- **To what degree would you delegate course registration to an AI assistant? Discuss with your neighbor(s).**
  - Finding required and elective courses
  - Collecting information about courses and instructors
  - Determining candidate courses
  - Checking schedule conflicts
  - Recommending courses for this semester
  - Making the final selection

Image Source: CIO.com

Key Takeaways

- **Technology alone does not guarantee better decisions**
- **Information must be structured and contextualized**
- **MIS provides the foundation for managerial judgment**

Next Class...

- **In the next class, we will cover:**
  - [Chapter 1] Information Systems in Global Business Today

Q & A